1. Understanding the Basics of Linear Algebra

Concepts to Learn:

- **Vectors and Scalars**: Learn the difference between vectors (e.g., a list of numbers) and scalars (a single number).
- **Dot Product**: Understand how to calculate the dot product of two vectors.
- **Matrix Operations**: Get familiar with matrix addition, multiplication, and transposition.

2. Basic Statistics and Probability

Concepts to Learn:

- **Mean, Median, Mode**: Understand these measures of central tendency.
- **Variance and Standard Deviation**: Learn how to measure the spread of data.
- **Correlation and Covariance**: Get an understanding of how variables relate to each other.

3. Understanding the Concept of a Line

Concepts to Learn:

- **Slope-Intercept Form**: Learn the equation of a line: $y = mx + cy$, where m is the slope and c is the intercept.
- **Plotting a Line**: Understand how to plot a straight line given its equation.

4. Introduction to Python and Numpy Basics

Concepts to Learn:

- **Python Basics**: Understand loops, functions, and basic data structures like lists and dictionaries.
- **NumPy Basics**: Learn how to create arrays, perform basic arithmetic operations, and use NumPy functions.

5. Simple Linear Regression by Hand

Concepts to Learn:

- **Linear Relationship**: Understand the relationship between the independent variable X and the dependent variable y.
- **Manual Calculation of Slope and Intercept**: Practice calculating these by hand using small data sets.

6. Plotting Data and Understanding Scatter Plots

Concepts to Learn:

- **Scatter Plots**: Learn how to visualize data points in a 2D plane.
- **Understanding Correlation**: Get a feel for positive, negative, and no correlation by looking at scatter plots.

7. Simple Gradient Descent Introduction

Concepts to Learn:

- **Gradient Descent**: Understand the basic idea behind minimizing a function.
- **Cost Function**: Learn about the cost function in linear regression (mean squared error).

Subject Code : BCSE0731: MACHINE LEARNING LAB

Lab (Day 2)

1. Given two vectors a = [1 , 3 , − 5] and b = [4 , − 2 , − 1], calculate the dot product of these vectors.

2. Given the dataset data = [2 , 4 , 4 , 4 , 5 , 5 , 7 , 9], calculate the mean, variance, and standard deviation of the dataset.

3. Plot the line with a slope m = 3 and intercept c = − 2 on a graph for x values ranging from -5 to 5.

4. Create a NumPy array with the elements [2 , 4 , 6 , 8 , 10]. Write a function to return an array with each element squared.

5. Given the dataset X = [1 , 2 , 3 , 4] and y = [2 , 2.5 , 3 , 3.5], manually calculate the slope m and intercept c of the best-fit line using the linear regression formulas.

6. Create a scatter plot for the following dataset: X = [1 , 2 , 3 , 4 , 5] and y = [2 , 3 , 5 , 7 , 11]. Comment on the relationship between X and y.

7. Implement a simple gradient descent algorithm to minimize the cost function for the dataset X = [1 , 2 , 3 , 4] and y = [2 , 4 , 6 , 8]. Start with an initial slope m = 0 and intercept c = 0, and run the algorithm for 100 iterations with a learning rate of 0.01.

Lab Question

Estimate parameters of a model based on Linear Regression method using a given set of training data set.